

VALIDATION OF RAINFALL ONSET DETECTION FORMULA USING CHIRPS DAILY RAINFALL DATA FOR IBADAN, NIGERIA (2021–2023)

Executive Summary

This report presents the validation of a rainfall onset detection formula using daily rainfall observations from the Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS) dataset. The analysis was conducted for Ibadan, Nigeria, over three years (2021–2023) to evaluate the consistency and reliability of the rainfall onset detection methodology.

The primary objective was to determine whether the proposed onset detection rule could consistently identify genuine rainfall onset dates suitable for agricultural planning and crop establishment.

1. Introduction

Rainfall onset is a critical parameter in agricultural planning, particularly for rain-fed farming systems. Accurate identification of rainfall onset enables farmers and decision-makers to determine appropriate planting dates, reduce the risk of crop failure, and improve agricultural productivity.

This study applies the proposed rainfall onset formula to daily rainfall observations extracted for Ibadan, Nigeria, and evaluates the validity of the detected onset dates using post-onset rainfall persistence and dry spell analysis.

2. Study Area

Location: Ibadan, Oyo State, Nigeria

Coordinates:

Latitude: 7.3775°N

Longitude: 3.9470°E

Ibadan is located within the humid tropical climate zone of southwestern Nigeria and experiences a distinct wet and dry season, making rainfall onset determination particularly important for agricultural activities.

3. Data Source

Dataset: CHIRPS Version 2.0 Daily Rainfall Dataset

Temporal Coverage:

- March–June 2021
- March–June 2022
- March–June 2023

Spatial Resolution:

- $0.05^{\circ} \times 0.05^{\circ}$

Data Type:

- Daily precipitation (mm/day)

4. Data Overview

Sample Records (First Five Observations)

	time	precip	latitude	longitude
0	2021-03-01	0.0	7.375	3.925003
1	2021-03-02	0.0	7.375	3.925003
2	2021-03-03	0.0	7.375	3.925003
3	2021-03-04	0.0	7.375	3.925003
4	2021-03-05	0.0	7.375	3.925003

Summary Statistics

Statistic	Value
Records	366
Mean Rainfall (mm)	4.285336
Maximum Rainfall (mm)	40.504288
Missing Values	0

Interpretation

The dataset consists of 366 daily rainfall observations covering the study period. The average daily rainfall was approximately 4.29 mm, while the maximum recorded daily rainfall event reached 40.50 mm. No missing values were detected, indicating a complete dataset suitable for analysis.

5. Methodology

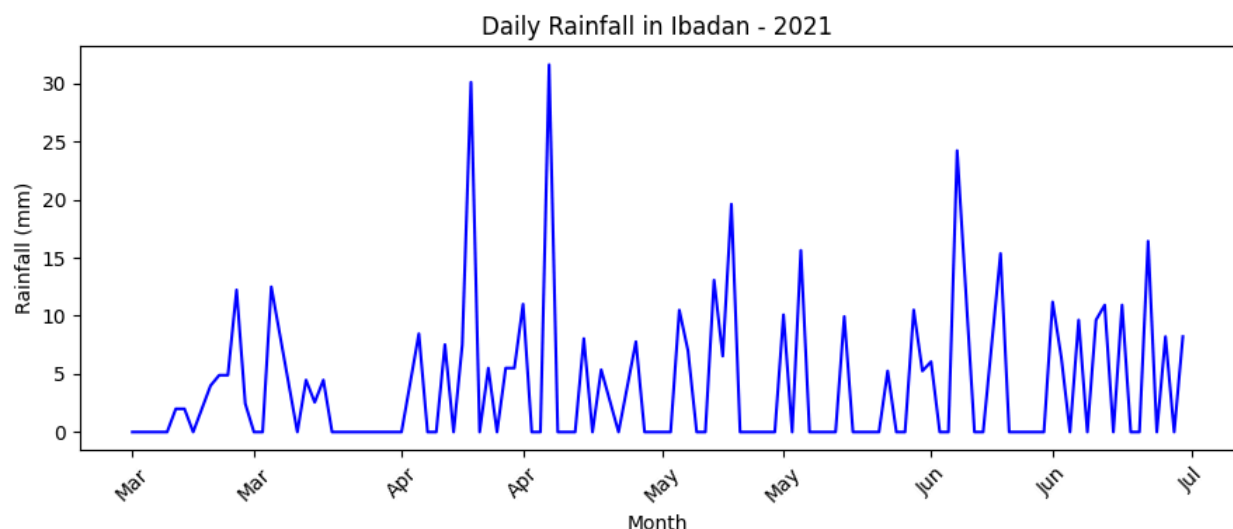
The rainfall onset detection formula identifies the beginning of the rainy season based on cumulative rainfall conditions over a short time window.

The methodology consists of:

1. Extraction of daily rainfall data.
2. Calculation of cumulative rainfall over consecutive days.
3. Identification of candidate onset dates.
4. Validation using post-onset rainfall persistence.
5. Detection of false onsets using dry spell analysis.

6. Annual Rainfall Characteristics

2021 Rainfall Distribution

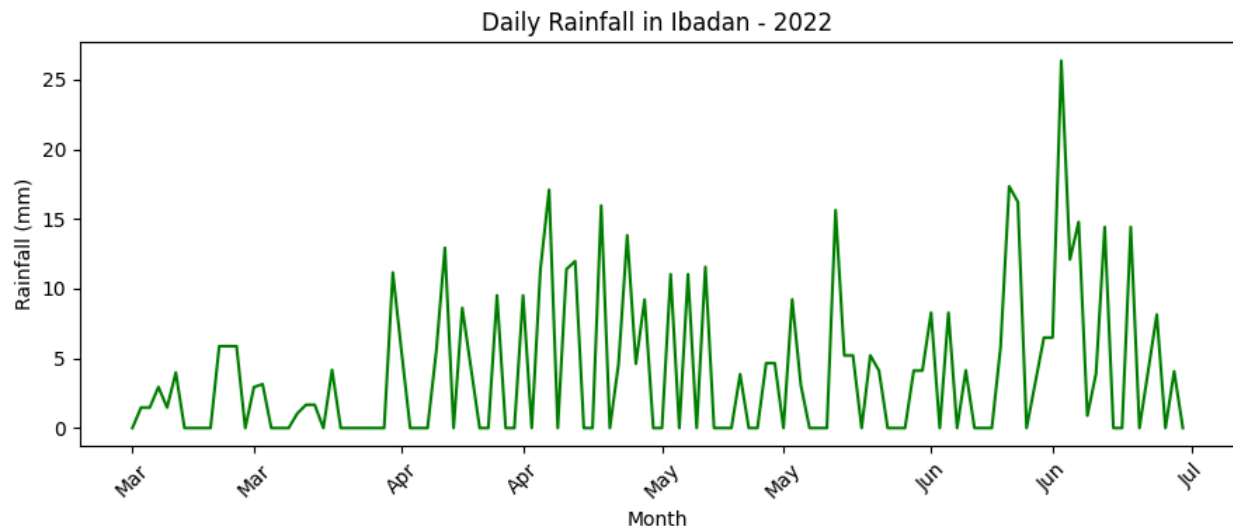


Observations

- Describe the seasonal rainfall pattern.
- Highlight rainfall build-up towards the onset period.

- Discuss rainfall intensity and variability.

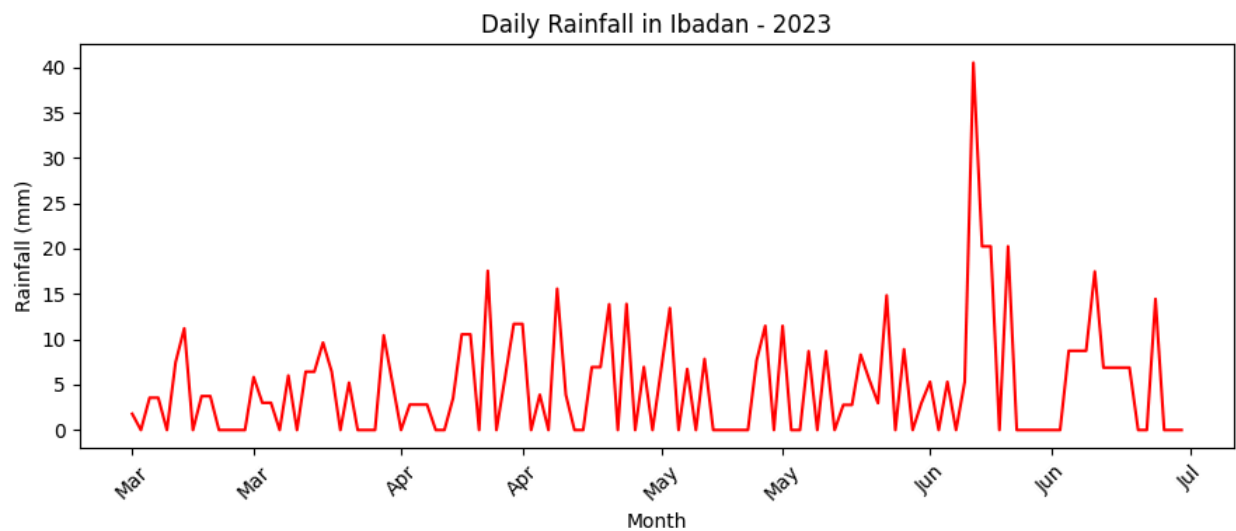
2022 Rainfall Distribution



Observations

- Describe rainfall progression.
- Identify wet and dry intervals.
- Discuss annual variability relative to 2021.

2023 Rainfall Distribution



Observations

- Describe the rainfall pattern.
- Highlight the onset period.
- Compare with previous years.

7. Rainfall Onset Detection Results

Detected Onset Dates

Year	Onset Date
2021	2021-03-13
2022	2022-04-08
2023	2023-03-23

Interpretation

The analysis detected rainfall onset in each of the three years examined. The earliest onset occurred during March 2021, while the latest onset was observed in April 2022. These variations demonstrate interannual rainfall variability typical of tropical climates.

8. Onset Validation Assessment

The detected onset dates were further evaluated using post-onset rainfall persistence and dry spell characteristics.

Year	Onset Date	Rain Next 10 Days (mm)	Max Dry Days	Genuine Onset?
2021	2021-03-13	46.700001	2	Yes
2022	2022-04-08	43.400002	2	Yes
2023	2023-03-23	37.000000	3	Yes

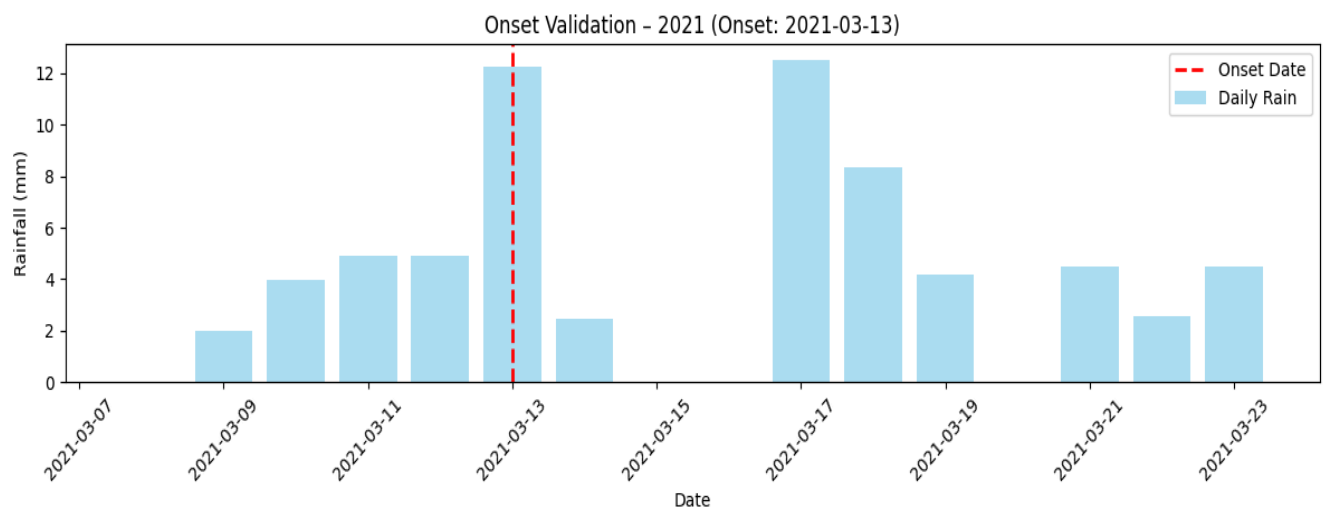
Interpretation

All detected onset dates were classified as genuine onsets. Following each onset event, substantial rainfall continued during the subsequent ten-day period, while dry spell lengths remained below critical thresholds. This provides strong evidence that the formula successfully distinguished true onset events from temporary rainfall occurrences.

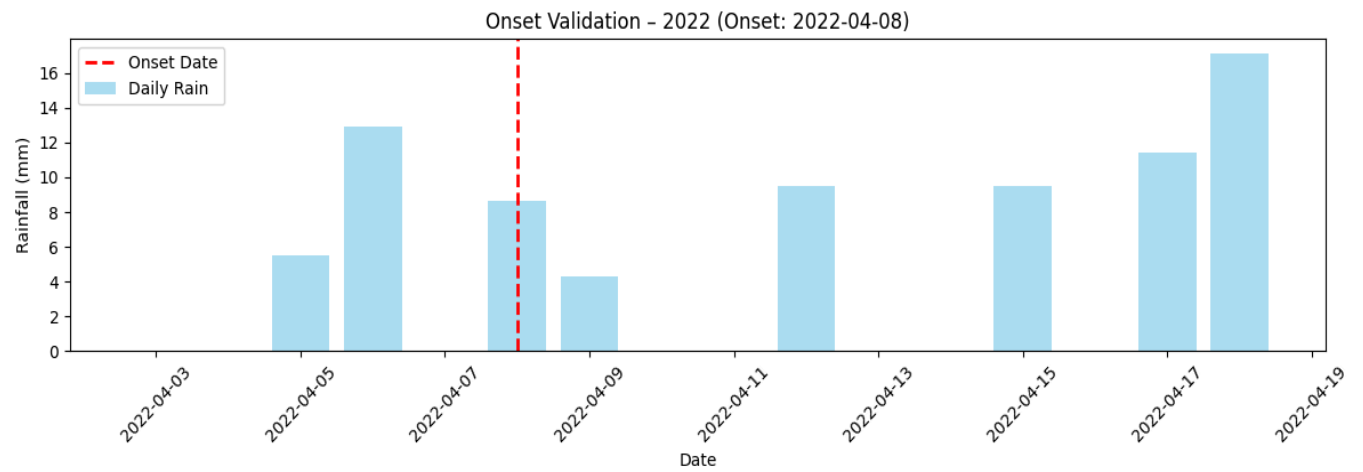
9. Onset Visualization

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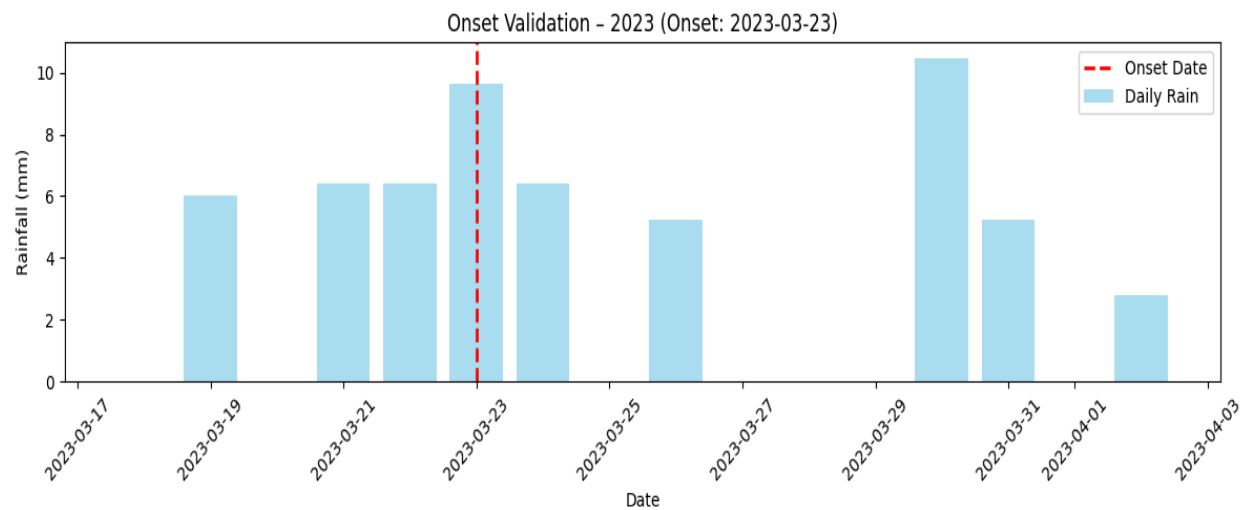
Onset Detection – 2021



Onset Detection – 2022



Onset Detection – 2023



10. Overall Validation Results

Metric	Value
Total Years Analysed	3
Genuine Onsets	3
False Onsets	0
Success Rate	100%

11. Key Findings

- The rainfall onset detection formula successfully identified an onset date for each year analysed.
- All detected onset dates passed the validation criteria.
- No false onset events were observed during the study period.
- The formula demonstrated strong consistency across multiple years.
- Post-onset rainfall persistence supports the reliability of the detected onset dates.
- The methodology shows potential for agricultural planning and crop establishment decision-making.

12. Conclusions

The rainfall onset detection formula was successfully validated using daily CHIRPS rainfall data for Ibadan, Nigeria, between 2021 and 2023.

The analysis achieved a 100% validation success rate, with all identified onset dates confirmed as genuine onset events. The results indicate that the methodology is suitable for identifying agriculturally relevant rainfall onset periods and may serve as a useful decision-support tool for rain-fed agricultural systems.

Further work may include testing alternative rainfall thresholds, expanding the temporal coverage, and evaluating performance across additional climatic regions.